

SME Risk Analytics: Rule-Based vs Machine Learning Model

Problem Statement

Small and medium enterprises (SMEs) often lack consistent and scalable methods for financial risk assessment. Traditional rule-based approaches are interpretable but limited in adaptability, while machine learning models offer predictive power but can lack transparency. This project explores how both approaches compare in classifying SME risk.

Approach

This project builds and compares two complementary risk assessment methods:

- **Rule-Based Model:** Financial indicators (liquidity, profitability, asset structure, and discretionary metrics) were engineered into a composite scoring framework to classify SMEs into Low, Medium, and High risk categories.
- **Machine Learning Model:** Supervised learning models (Logistic Regression and Random Forest) were trained on the same financial features to predict SME risk categories.
- **Evaluation:** Both approaches were compared using classification alignment to assess consistency between rule-based logic and ML predictions.

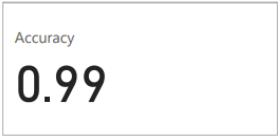
Tools Used

- Python (Pandas, Scikit-learn)
- Google Colab
- Power BI
- Excel (data preparation)

Key Result

The machine learning model showed strong alignment with the rule-based framework, achieving approximately 0.99 classification agreement across SME risk categories. This suggests that structured financial indicators contain strong predictive signals and can be effectively modelled using both traditional and ML approaches.

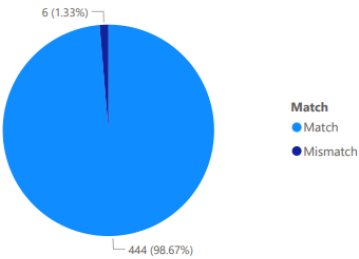
Model Comparison & Performance



Model agreement between rule-based and ML predictions

FQ1	risk_category	Predicted_Risk	Match
YES	High Risk	High	Match
YES	High Risk	Medium	Misma
YES	Low Risk	Low	Match
YES	Low Risk	Medium	Misma
YES	Medium Risk	High	Misma
YES	Medium Risk	Low	Misma
YES	Medium Risk	Medium	Match

Count of Match by Match



Count of Match by Match

